NP-hardness of approximating constant-round total Boolean protocol leaf count

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Editorial note. Faithful editable transcription from the official CCC2021 CC BY4.0 published version, article31, 2021-07-08. Theorem4 and its formal Corollary24 conclusion are one constant-round deterministic protocol-size hardness result. Full three-round Boolean-matrix construction, chromatic-number gap prerequisite, large-alphabet warmup, cost characterization, deterministic fooling-set round elimination and approximation reduction proofs are retained with original numbered statements. Original total Boolean NxN input, constant round/approximation conditions, Alice/Bob-first conventions and model definitions are retained. Original mathematical expressions and reduction-time wording are transcribed without author-proof correction or added assumptions. The complementary leaf-set overlines were read from official p31:18; typography-only line breaks preserve all terms. All65 official bibliography entries retained. Randomized/conditional extensions in Sections6–7 are not claimed by this deterministic item; the full official original remains available. Supporting statements and direct consequences are not counted separately.

Allowed standard prerequisites (editorial).

- Chromatic-number hardness of approximation. Exact Theorem9 with original quantifiers and references[42,24,20,65] retained.
- Explicit universal sets. Exact Definition15 and Lemma16[51], including size/time parameters, retained; no recursion into prior proof.

Theorem 4. *For any $d \geq 3$ there exists a constant $\Delta_d > 0$ such that the following holds. If there exists a polynomial-time algorithm which, when given a total two-player Boolean-valued function $f : [N] \times [N] \to \{0,1\}$ represented as a Boolean matrix of dimensions $N \times N$, approximates $L_d^A(f)$ within a factor of $1 + \Delta_d$, then* P = NP.

Author preliminaries and communication model (Section 2)

For a positive integer n, we let $[n]$ denote the set $\{1, \ldots, n\}$.

Binomial Coefficients and Projections. The binomial coefficient $\binom{n}{k}$ equals the number of distinct subsets of $[n]$ with exactly k elements. Similarly, $\binom{n}{\leq k}$ denotes the number of distinct subsets of $[n]$ that have at most k elements. Finally, $\binom{[n]}{k}$ denotes the set of all subsets of $[n]$ with exactly k elements. If $x \in \{0,1\}^n$ and $S = \{i_1, \ldots, i_k\} \in \binom{[n]}{k}$, then $x_S \in \{0,1\}^k$ denotes the projection of x to S, given by $x_S = x_{i_1} \cdots x_{i_k}$.

Communication complexity. We assume basic familiarity with communication complexity [38]. We write the definitions here for clarity.

Definition 5 (Protocol)**.** *Let $\mathcal{A}, \mathcal{B}, \mathcal{Z}$ be finite sets. A deterministic protocol π over $\mathcal{A} \times \mathcal{B} \times \mathcal{Z}$ is a rooted tree:*

- *Each node v is associated with a rectangle $\pi^{-1}(v) = A \times B$, with $A \subseteq X$ and $B \subseteq Y$.*
- *Each non-leaf node v, associated with a rectangle $\pi^{-1}(v) = A \times B$, is labeled by either (a) a partition $A = \dot{\bigcup}_{c \in P_v} A_c$ of A, in which case we say it is Alice's node or (b) a partition $B = \dot{\bigcup}_{c \in P_v} B_c$ of B, in which case we say it is Bob's node.*
- *Each leaf node is labeled by an element of the output domain $\mathcal{Z}$.*
- *The rectangle associated with the root is the input domain $\mathcal{A} \times \mathcal{B}$.*
- *If a non-leaf node v of Alice is associated with rectangle $\pi^{-1}(v) = A \times B$ and (a) labeled by a partition $A = \dot{\bigcup}_{c \in P_v} A_c$ of A, then for each $c \in P_v$ there will be one child v_c of v, which will be associated with the rectangle $\pi^{-1}(v_c) = A_c \times B$; similarly for Bob's nodes.*

We let the leaf complexity of π, written $L(\pi)$, be the number of leaf nodes of π. We let the round complexity of π, written $R(\pi)$, be the height of π, i.e., the maximum number of edges in any root-to-leaf path of π. A root-to-leaf path $v_1 \to \cdots \to v_{k+1}$ is said to have communication length $\sum_{i=1}^{k} \lceil \log |P_{v_i}| \rceil$ (where $|P_{v_i}|$, as defined above, is the number of parts in the partition of A or B associated with node v_i). We then let the communication complexity of π, written $C(\pi)$, be the maximum communication length of any root-to-leaf path of π. Given $(a, b) \in \mathcal{A} \times \mathcal{B}$, we let $\pi(a, b)$ denote the (unique) leaf v of π having $(a, b) \in \pi^{-1}(v)$. For $z \in \mathcal{Z}$, we may write $\pi(a, b) = z$ to mean that the leaf $\pi(a, b)$ is labeled by z. A randomized protocol over $\mathcal{A} \times \mathcal{B} \times \mathcal{Z}$ is a distribution over deterministic protocols over $\mathcal{A} \times \mathcal{B} \times \mathcal{Z}$. We will use a boldface Greek letter, such as $\boldsymbol{\pi}$, to denote a protocol sampled from this distribution. We then let $L(\boldsymbol{\pi})$ be the maximum $L(\pi)$ over all π in the support of $\boldsymbol{\pi}$, and likewise for $R(\boldsymbol{\pi})$ and $C(\boldsymbol{\pi})$.

Definition 6. *A function $f : \mathcal{A} \times \mathcal{B} \to \mathcal{Z}$ is said to be computed by a deterministic protocol π over $\mathcal{A} \times \mathcal{B} \times \mathcal{Z}$ if we have $f(a, b) = \pi(a, b)$ for every $(a, b) \in \mathcal{A} \times \mathcal{B}$. Furthermore, if $\varepsilon \in [0, 1]$, then f is said to be computed with error ε by a randomized protocol $\boldsymbol{\pi}$ if, for every $(a, b) \in \mathcal{A} \times \mathcal{B}$, $\Pr[\boldsymbol{\pi}(a, b) = f(a, b)] > 1 - \varepsilon$ (the probability is over the choice of $\boldsymbol{\pi}$). We may then define:*

- $L^A_d(f)$ *is the minimum leaf complexity* $L(\pi)$ *among all deterministic protocols* π *that compute* f*, and have round complexity* $R(\pi) \leq d$*, and such that the root node of* π *is Alice's.* $L^B_d(f)$ *is defined likewise, but for protocols where the root node is Bob's.*
- $L^A_{d,\varepsilon}(f)$ *is the minimum* $L(\boldsymbol{\pi})$ *among all randomized protocols* $\boldsymbol{\pi}$ *that compute* f *with error* ε*, and have* $R(\boldsymbol{\pi}) \leq d$*, and such that the root node of* $\boldsymbol{\pi}$ *is (always) Alice's.* $L^B_{d,\varepsilon}(f)$ *is defined likewise for randomized protocols where the root node is (always) Bob's.*
- $C^A_d, C^B_d, C^A_{d,\varepsilon}, C^B_{d,\varepsilon}$ *are defined analogously where the communication complexity* $C(\pi)$ *replaces the leaf complexity* $L(\pi)$*.*

Since the communication transcript of a given run of protocol determines the leaf, it must follow that:

Proposition 7. *For any protocol* π*,* $\log L(\pi) \leq C(\pi)$*.*

Chromatic number. All our NP-hardness reductions are from the chromatic number problem:

Definition 8 (Chromatic number)**.** *A coloring of an undirected graph* G*, is a partition of the vertices such that no edge has both endpoints in the same part. The chromatic number of a graph* G*, denoted* $\chi(G)$*, is the smallest number of parts in a coloring of* G*.*

The NP-hardness of approximating the chromatic number has been established by a series of results [42, 24, 20], culminating in a paper by Zuckerman [65], where the following was proven:

Theorem 9 (Hardness For Chromatic Number)**.** *For every* $\epsilon > 0$*, there is a deterministic polynomial time algorithm that on an input* $x \in \{0,1\}^*$ *outputs a graph* G *on* n *vertices such that*

- *if* x *is a YES instance of SAT, then* $\chi(G) \leq n^\epsilon$*, and*
- *if* x *is a NO instance of SAT, then* $\chi(G) \geq n^{1-\epsilon}$*.*

Author three-round cost characterization (Section 3)

We start by observing that deterministic 3-round communication complexity may be approximated by a very simple combinatorial quantity.

Definition 10. *Let* A, B, Z *be finite sets, and let* M *be an* $A \times B$ *matrix over* Z*. Let* $\mathcal{P} = \{P_i\}_{i\in[k]}$ *be a partition of (the columns)* B *for some* $k \in \mathbb{N}$*. (That is,* $\emptyset \neq P_i \subseteq B$*,* $\bigcup_{i\in[k]} P_i = B$ *and* $P_i \cap P_j = \emptyset$ *for every* $i \neq j$*.) For a subset* $P \subseteq B$ *of columns, we denote by* $\mathrm{Cost}_M(P)$ *the number of distinct rows of* M *restricted to columns in* P*, i.e.,*

$$\mathrm{Cost}_M(P) = |\{x_P \in Z^P \mid x \in Z^B \text{ is a row of } M\}|.$$

We further define $\mathrm{Cost}_M(\mathcal{P})$ *to be* $\sum_{i=1}^k \mathrm{Cost}_M(P_i)$*.*

Proposition 11. *Let $f : A \times B \to Z$ be a function and M be the $A \times B$ communication matrix (with entries in Z) that corresponds to f. Let $q \in \mathbb{N}$ be the maximum number of distinct values in a single row of M. We then have*

$$L \leq L_3^B(f) \leq L \cdot q, \qquad \text{where } L = \min_{\mathcal{P}} \mathrm{Cost}_M(\mathcal{P}),$$

and where the minimum is taken over all partitions $\mathcal{P}$ of B. Furthermore we have that

$$\log L \leq C_3^B(f) \leq \log L + \log q + O(1).$$

Proof. It may be easily seen that $L_2^A(f)$ is lower-bounded by the number of distinct rows in the communication matrix of f. Because if Alice's partition of the rows includes a part with two different rows, then Bob's ensuing partition of the columns cannot avoid having a non-monochromatic column. It then follows that, if we have a 3-round protocol π where Bob speaks first, and $\mathcal{P}$ is Bob's partition of the columns in the first round, then $\mathrm{Cost}_M(\mathcal{P})$ is a lower-bound on smallest number of leaves that π needs to use to compute f, and thus $L_3^B(f) \geq L$. The lower-bound on $C_3^B(f)$ now follows from Proposition 7. Conversely, it is also easy to see that $L_2^A(f)$ is upper-bounded by q times the number of distinct rows in the communication matrix of f. The protocol that achieves this bound has Alice tell which kind of row she has, and now in each rectangle the rows are all equal, hence Bob can just tell Alice the color of his column in this row, of which there are q possibilities. Thus $L_3^B(f) \leq L \cdot q$, and the same protocol shows that $C_3^B(f) \leq \log L + \log q + O(1)$. □

In general, the approximate factor q of Proposition 11 can be as large as $|Z|$. However, in the following construction, we will construct a matrix where each row has at most $q = 3$ distinct values; in this case, Proposition 11 guarantees that $\min_{\mathcal{P}} \mathrm{Cost}_M(\mathcal{P})$ provides a 3-factor approximation of $L_3(M)$.

Theorem 12. *Given an undirected graph $G = ([n], E)$ with n vertices and $|E| = m > 0$ edges, one may construct in deterministic polynomial time a function $f_G : [a] \times [n] \to \{0, 1, \ldots, \ell\}$, with $\ell = m^2 n$, $k = \sqrt{n\ell} = mn$ and $a = \ell + m \cdot k^2$, such that*

$$\chi(G) \cdot \ell \leq L_3^B(f_G) \leq \chi(G) \cdot 6\ell.$$

Furthermore, we also have

$$\log \chi(G) + \log \ell \leq C_3^B(f_G) \leq \log \chi(G) + \log \ell + O(1).$$

Proof. We let A_ℓ denote the $\ell \times 1$ column vector,

$$A_\ell = \begin{bmatrix} 1 \\ \vdots \\ \ell \end{bmatrix}.$$

We let B_k and C_k denote the $k^2 \times 1$ column vectors

$$B_k = \begin{bmatrix} 1 \\ \vdots \\ 1 \\ \vdots \\ k \\ \vdots \\ k \end{bmatrix}, \qquad C_k = \begin{bmatrix} 1 \\ \vdots \\ k \\ \vdots \\ 1 \\ \vdots \\ k \end{bmatrix},$$

where each value $i \in [k]$ appears k times. Given an edge $\{v, w\} \in [n] \times [n]$, with $v < w$, we define the $k^2 \times n$ matrix M so that

$$M_{\{v,w\}} = \begin{bmatrix} 0 & \cdots & 0 & B_k & 0 & \cdots & 0 & C_k & 0 & \cdots & 0 \end{bmatrix},$$

where B_k appears in the v-th column and C_k in the w-th column. Finally, let $E = \{e_1, \ldots, e_m\}$ denote the edges of G; then we define the $a \times n$ communication matrix f_G so that

$$f_G = \begin{bmatrix} A_\ell & \cdots & A_\ell \\ & M_{e_1} & \\ & \vdots & \\ & M_{e_m} & \end{bmatrix} \in \{0, 1, \ldots, \ell\}^{a \times n},$$

where $a = \ell + m \cdot k^2$. Observe that each row of f_G has at most $q = 3$ distinct values; thus, Proposition 11 provides a 3-factor approximation. We now show the stated inequality, namely, that

$$\chi(G) \cdot \ell \leq L_3^B(f_G) \leq \chi(G) \cdot 2q\ell.$$

The upper-bound is easy to see. Given a coloring of G into $\chi(G)$ colors, we may take the 3-round protocol π where Bob first tells Alice which color his vertex has. This partitions the columns by a partition $\mathcal{P} = \{P_c\}_{c \in [\chi(G)]}$ formed of the various color classes of our coloring. Fix any color c and consider $\mathrm{Cost}_{f_G}(P_c)$. Since P_c is an independent set of G, the C_k and B_k columns of each M_{e_i} submatrix will always be placed in different parts; therefore, $\mathrm{Cost}_{f_G}(P_c) \leq \ell + mk$, where the first term counts the number of rows in A_ℓ and the second term counts the number of distinct rows in M_{e_i} restricted to columns of P_c for each $i \in [m]$. Using Proposition 11, we obtain

$$L_3^B(f_G)/q \leq \mathrm{Cost}_{f_G}(\mathcal{P}) = \sum_c \mathrm{Cost}_{f_G}(P_c) \leq \chi(G) \cdot (\ell + mk) = \chi(G) \cdot 2\ell,$$

and Proposition 11 also gives us

$$C_3^B(f_G) \leq \log \chi(G) + \log \ell + O(1).$$

For the other direction, let π be any 3-round protocol for f_G. The first round of π partitions the columns of f_G, which is to say, it partitions the vertices of G by some partition $\mathcal{P} = \{P_i\}_{i \in I}$. We claim that $\chi(G) \cdot \ell \leq \mathrm{Cost}_{f_G}(\mathcal{P})$ by analyzing the following two cases.

1. If the partition $\mathcal{P}$ does not form a coloring of G, then there must exist an edge $\{v, w\}$ such that the v-th column and the w-th column of G are placed in the same part P_i. This means that the C_k and B_k columns of the $M_{\{v,w\}}$ sub-matrix are both placed in P_i. In this case, we have $\mathrm{Cost}_{f_G}(P_i) \geq k^2 \geq n \cdot \ell \geq \chi(G) \cdot \ell$, and thus the lower bound holds.

2. Otherwise, suppose that the partition $\mathcal{P}$ does form a coloring of G. Then the number of parts is $\geq \chi(G)$, and so just the contribution from the first ℓ rows gives us $\mathrm{Cost}_{f_G}(\mathcal{P}) \geq \chi(G) \cdot \ell$.

The lower-bounds on $L_3^B(f_G)$ and $C_3^B(f_G)$ then follow from Proposition 11. □

The following corollary shows that it is NP-hard to approximate $L_3^B(f)$, for a given total two player function $f : [N] \times [N] \to [N]$, with an approximation ratio better than (roughly) $N^{1/5}$.

Corollary 13. *For every $L \subseteq \{0,1\}^*$ in* NP *and every constant $\varepsilon > 0$, there exists a polynomial-time algorithm that, on input $x \in \{0,1\}^*$, outputs a communication matrix $M \in [N]^{N \times N}$ such that if $x \in L$ then $L_3^B(M) \leq N$, and if $x \notin L$ then $L_3^B(M) > N^{6/5-\varepsilon}$.*

Proof. We compose the hardness of approximation result for chromatic number in Theorem 9 with the reduction of Theorem 12, padding the communication matrix with all-0 columns to make it square (since Bob speaks first, this adds at most one leaf), so the communication matrix of f is an $N \times N$ matrix with $N = a = \Theta(n^5)$. The parameters then come from the fact that $n = \Theta(N^{1/5})$. □

Using the bounds on C_3^B instead of the bounds on L_3^B from Theorem 12, we conclude that it is NP-hard to approximate $C_3^B(f)$, for a given total two player function $f : [N] \times [N] \to [N]$, with an approximation ratio better than (roughly) 6/5. More precisely:

Corollary 14. *For every $L \subseteq \{0,1\}^*$ in* NP *and every $\varepsilon > 0$, there exists a polynomial-time algorithm that, on input $x \in \{0,1\}^*$, outputs a communication matrix $M \in [N]^{N \times N}$ such that if $x \in L$ then $C_3^B(M) \leq \log N$, and if $x \notin L$ then $C_3^B(M) > (\frac{6}{5} - \varepsilon) \log N$.*

Hardness for deterministic 3-round protocols (author Section 4)

Building on the proof ideas presented in Section 3, in this section, we prove NP-hardness of approximating the communication complexity of deterministic 3-round protocols. A key building block is to use the notion of universal set.

Definition 15 (Universal set)**.** *Let $r, c, k \in \mathbb{N}$, and let $M \in \{0,1\}^{r \times c}$ be a matrix whose columns are $M^{(1)}, \ldots, M^{(c)}$. We say that M is (c,k)-universal if, for every set $\{y_1, \ldots, y_k\} \subseteq [c]$ of k columns of M, the matrix*

$$\begin{bmatrix} M^{(y_1)} & \cdots & M^{(y_k)} \end{bmatrix}$$

has 2^k distinct rows. The set of all the rows of M is called a (c,k)-universal set.

We use the explicit construction of a universal set due to [49].

Lemma 16 (Naor, Schulman and Srinivasan [51])**.** *There exists a deterministic algorithm that, given c and $k \in \mathbb{N}$, outputs a (c,k)-universal matrix $M \in \{0,1\}^{r \times c}$ such that $r = 2^{k+O(\log^2 k)} \cdot \log c$ in time a polynomial in c and 2^k.*

Now we state the main result of this section.

Theorem 17. *Let $\epsilon > 0$ be an arbitrary constant. Given an undirected graph $G = ([n], E)$ with n vertices and $|E| = m > 0$ edges, one may construct in deterministic polynomial time a function $f_G : [a] \times [b] \to \{0,1\}$ and a number $\ell \in \mathbb{N}$, with $a, b, \ell = O(n^8)$, such that*

$$\chi(G) \cdot \ell \leq L_3^B(f_G) \leq \chi(G) \cdot \ell^{1+\epsilon}$$

and

$$\log \chi(G) + \log \ell \leq C_3^B(f_G) \leq \log \chi(G) + (1+\epsilon) \cdot \log \ell.$$

The idea of the proof is to build upon the reduction in the proof of Theorem 12, by replacing each column with entries from $[\ell]$ with a block of columns that have entries from $\{0,1\}$. The difficulty in making this work is that a partition of the columns might not respect our blocks and could place columns from the same block into different parts. We solve this by thinking as follows. Either the partition is large, meaning it has many parts, so the protocol also has many leaves, which proves our lower bound, or otherwise for any block C_v of columns the part P_i which has most columns from C_v has many columns from C_v; we may then act as if "C_v was placed in P_i".

Proof. Let $t, k, c \in \mathbb{N}$ be parameters chosen later. Let $A \in \{0,1\}^{r \times c}$ and $M \in \{0,1\}^{s \times c}$ be the (c,t)-universal and (c,k)-universal matrices, respectively, that are constructed by the polynomial-time deterministic algorithm of Lemma 16; then we have $r = 2^{(1+o(1)) \cdot t} \cdot \log c$ and $s = 2^{(1+o(1)) \cdot k} \cdot \log c$. Let $x_1, \ldots, x_s$ be the rows of M. We then let B and C denote the $s^2 \times c$ matrices

$$B = \begin{bmatrix} x_1 \\ \vdots \\ x_1 \\ \vdots \\ x_s \\ \vdots \\ x_s \end{bmatrix}, \qquad C = \begin{bmatrix} x_1 \\ \vdots \\ x_s \\ \vdots \\ x_1 \\ \vdots \\ x_s \end{bmatrix},$$

where a row x_i appears s times for each $i \in [s]$. Given an edge $\{v, w\} \in [n] \times [n]$, with $v < w$, we define the $s^2 \times c \cdot n$ matrix:

$$M_{\{v,w\}} = \begin{bmatrix} 0 & \cdots & 0 & B & 0 & \cdots & 0 & C & 0 & \cdots & 0 \end{bmatrix},$$

where B appears in the v-th block of c columns and C in the w-th block of c columns. Finally, let $E = \{e_1, \ldots, e_m\}$ denote the edges of G; then we define

the $(r + m \cdot s^2) \times c \cdot n$ communication matrix f_G so that

$$f_G = \begin{bmatrix} A & \cdots & A \\ & M_{e_1} & \\ & \vdots & \\ & M_{e_m} & \end{bmatrix} \in \{0,1\}^{(r+m\cdot s^2)\times c\cdot n}.$$

Let $\ell := 2^t$. Let $\mathcal{P}$ be a partition of the columns of f_G that minimizes $\mathrm{Cost}_{f_G}(\mathcal{P})$. We now show that, for a suitable choice of t, k and c,

$$\chi(G) \cdot \ell \leq \mathrm{Cost}_{f_G}(\mathcal{P}) \leq \chi(G) \cdot \ell^{1+o(1)}.$$

This will complete the proof, because Proposition 11 shows that, for a binary matrix f_G, $\mathrm{Cost}_{f_G}(\mathcal{P})$ is a 2-factor approximation of $L_3^B(f_G)$ and $\log \mathrm{Cost}_{f_G}(\mathcal{P})$ is an approximation of $C_3^B(f_G)$ up to an additive $O(1)$ term. We will choose the parameters t, k and c so that the following conditions are satisfied.

1. Condition 1. $r + ms \leq \ell^{1+o(1)}$.
2. Condition 2. $c \geq n\ell \cdot \max\{t, k\}$.
3. Condition 3. $2^{2k} \geq n\ell$.

Given that Condition 1 is satisfied, the complexity upper-bounds are easy to see. Let $\mathcal{P}$ be a coloring of G that partitions the vertex set into $\chi(G)$ parts. Since no class contains an edge, the C and B sub-matrices of each M_{e_i} sub-matrix will always be placed in different parts, and we thus have

$$\mathrm{Cost}(\mathcal{P}) \leq \chi(G) \cdot (r + ms) \leq \chi(G) \cdot \ell^{1+o(1)}.$$

For the other direction, we will lower-bound $\mathrm{Cost}(\mathcal{P})$ for every partition $\mathcal{P}$. To begin, if the number of parts is $|I| \geq n\ell$, then we must conclude that $\mathrm{Cost}(\mathcal{P}) \geq n\ell \geq \chi(G) \cdot \ell$, as desired. Otherwise, $|I| \leq n\ell$. Then for every block of columns $v \in [n]$ there must exist a part $P_{i(v)}$ which contains at least $c/|I| \geq c/n\ell$ columns from the v-th block. Observe that $c/n\ell \geq \max\{t, k\}$ by Condition 2. Now, either the mapping $v \mapsto i(v)$ is a valid coloring of G or not. First, suppose that the mapping $v \mapsto i(v)$ is not a valid coloring of G, meaning that there exists an edge $\{v, w\} \in E$ such that $i = i(v) = i(w)$. Then the v-th column block and the w-th column block each have at least $\max\{t, k\}$ columns in the same part P_i. This will mean that the C and B sub-matrices of the $M_{\{v,w\}}$ sub-matrix each have at least k columns in P_i. But then, since M is (c, k)-universal, it follows from Condition 3 that $\mathrm{Cost}(\mathcal{P}) \geq \mathrm{Cost}(P_i) \geq 2^{2k} \geq n\ell$. Next, suppose that $i : [n] \to \mathcal{P}$ does form a coloring of G. Then there exist at least $\chi(G)$ parts P_i each receiving at least t columns from some column block, and so, since A is (c, t)-universal, the contribution from the A columns give us $\mathrm{Cost}(\mathcal{P}) \geq \chi(G) \cdot 2^t = \chi(G) \cdot \ell$. This will give us the theorem, and all we are left to do is ensure that the various conditions can be met: We define t so that $\ell = 2^t$ satisfies that $\ell \leq (nm^2)^{1+\epsilon/2} < 2\ell$. To meet Condition 3, let k be the smallest integer satisfying that $n\ell \leq 2^{2k}$. Let $c := n\ell \cdot \max\{t, k\}$ so that Condition 2 is satisfied. Finally, observe that Condition 1 is satisfied because

$$\begin{aligned} r + ms &\leq \ell^{1+o(1)} \cdot \log c + m \cdot 2^{(1+o(1))\cdot k} \cdot \log c \\ &\leq \ell^{1+o(1)} + m(n\ell)^{\frac{1}{2}+o(1)} \cdot \log c \leq \ell^{1+\epsilon}, \end{aligned}$$

where the last inequality holds for all large n, m. □

We can now prove that L_3^A and C_3^A are hard to approximate, also for Boolean functions.

Theorem 18. *For every constant $\varepsilon > 0$, there exists a deterministic quasipolynomial-time algorithm that, on input $x \in \{0,1\}^*$, outputs a communication matrix $M \in \{0,1\}^{N \times N}$ and a number $k \in \mathbb{N}$, with $k \leq N = |x|^{O(1)}$, such that*

1. *if x is a YES instance of SAT, then $L_3^B(M) \leq k$ and $C_3^B(M) \leq \log k$, and*
2. *if x is a NO instance of SAT, then $L_3^B(M) \geq N^{\frac{1}{8}-\varepsilon} \cdot k$ and $C_3^B(M) \geq \log k + (\frac{1}{8} - \varepsilon) \log N$.*

Proof. We combine the two reductions of Theorems 9 and 17, which we invoke with the same parameter ε. Fix any input x and let G be an n-vertex graph that is produced by the reduction of Theorem 9 on input x. Let $M \in \{0,1\}^{a \times b}$ be the communication matrix of f_G, and $\ell \in \mathbb{N}$, be given by the reduction of Theorem 17 on input G, and set $N = \max(a,b) = O(n^8)$, $k = n^{\varepsilon} \cdot \ell^{1+\varepsilon}$. We may assume that $a = b$ without loss of generality, since otherwise we may pad M with all-0 rows or all-0 columns, and this changes the leaf complexity by at most a factor of 2, and the communication complexity by at most 1 bit, so this change makes no difference to the result. We verify that the inequalities are satisfied for M: If $x \in L$, then we have $L_3^B(M) \leq \chi(G) \cdot \ell^{1+\epsilon} \leq n^{\varepsilon} \cdot \ell^{1+\epsilon} = k$. If $x \notin L$, then we have

$$L_3^B(M) \geq \chi(G) \cdot \ell \geq n^{1-\epsilon} \cdot \ell \geq N^{\frac{1}{8}-O(\epsilon)} \cdot k.$$

Similar inequalities hold for $C_3^B(M)$. Since ε can be arbitrarily small, we may ignore the constant factors. □

From 3-rounds to multiple rounds using deterministic round elimination (author Section 5)

In this section we show how we can use an oracle for computing L_{d+1}^B in order to compute L_d^A. The gadget in our reduction involves a special function, XorEq_k, which is a small modification of the standard Equality function.

Definition 19. *The function* $\mathrm{XorEq}_k : [k] \times ([k] \times \{0,1\}) \to \{0,1\}$ *is given by*

$$\mathrm{XorEq}_k(x; y, z) = \begin{cases} z & \text{if } x \neq y, \\ 1 - z & \text{if } x = y. \end{cases}$$

The key property of XorEq_k is that it has a fooling set lower bound that is tight. In particular, $\{(x; y, z) : x = y\}$ is a fooling set of cardinality $2k$, and there is a 2-round protocol for solving XorEq_k with $2k$ leaves (where Alice just sends her full input to Bob, and he replies with the output).

Lemma 20 (Fooling set lower-bound for XorEq_k)**.** *Let π be a protocol for solving the function $f : ([k] \times [a]) \times ([k] \times \{0,1\}) \to \{0,1\}$ given by $f(x_0, x_1; y, z) = \mathrm{XorEq}_k(x_0; y, z)$.*[1] *Then*

$$\pi(x_0, x_1; y, z) \neq \pi(x_0', x_1'; y', z')$$

[1] In our definition of f, the input x_1 does not affect the output of the function. The fact that this lemma holds even when there is an extraneous input like x_1 will be used later.

if either

- $y = x_0 \neq x_0'$, *or*
- $y = x_0 = x_0'$ *and* $z \neq z'$.

Proof. First, suppose for contradiction that $\pi(x_0, x_1; y, z) = \pi(x_0', x_1'; y', z')$ and $y = x_0 \neq x_0'$. Since leaves are combinatorial rectangles, we can infer that $\pi(x_0', x_1'; y, z) = \pi(x_0, x_1; y, z)$. But since $y = x_0 \neq x_0'$, we know that

$$f(x_0, x_1; y, z) = 1 - z \neq z = f(x_0', x_1'; y, z)$$

so this contradicts that $\pi(x_0, x_1; y, z)$ is a monochromatic leaf. Similarly, if $y = x_0 = x_0'$ and $z \neq z'$, then we have $f(x_0, x_1; y, z) \neq f(x_0', x_1'; y', z')$, so $\pi(x_0, x_1; y, z) \neq \pi(x_0', x_1'; y', z')$ by monochromaticness. □

The main technical portion of our reduction is the following deterministic variant of the round-elimination lemma.

Lemma 21 (Restatement of Lemma 3)**.** *Let $d \geq 3$. Let $f : [a] \times [b] \to \{0,1\}$. Let $F : ([k] \times [a]) \times ([k] \times [b] \times \{0,1\} \times \{0,1\}) \to \{0,1\}$ be given by*

$$F(x_0, x_1; y_0, y_1, z, i) = \begin{cases} \mathrm{XorEq}_k(x_0; y_0, z) & \text{if } i = 0, \\ f(x_1; y_1) & \text{if } i = 1. \end{cases}$$

Then we have

$$\min\{4k, 2k - 2 + L_d^B(f)\} \leq L_{d+1}^A(F) \leq 2k + L_d^B(f).$$

Proof. The upper bound comes from the protocol where Alice skips the first round of communication, Bob sends i to Alice and begins running the best d-round Bob-first protocol for XorEq_k or f, based on whether $i = 0$ or $i = 1$. In particular, we have that

$$L_{d+1}^A(F) \leq L_d^B(\mathrm{XorEq}_k) + L_d^B(f) \leq 2k + L_d^B(f),$$

where the last upper bound uses that $d \geq 3$. We now argue the lower bound. Suppose π is a $(d+1)$-round Alice-first protocol for F. Let $\mathcal{L} = \{\ell_1, \ldots, \ell_{L(\pi)}\}$ denote the set of leaves of π. Our arguments split into two cases depending on whether there is a good input $x_0 \in [k]$. We say an input $x_0 \in [k]$ is good if all of Alice's inputs that begin with x_0 are placed in a single partition. More formally, let $\mathcal{P} = \{P_1, \ldots, P_{|\mathcal{P}|}\}$ be the partition of Alice's inputs corresponding to the first round of π. We say $x_0 \in [k]$ is good if there exists a $q \in [|\mathcal{P}|]$ such that $\{x_0\} \times [a] \subseteq P_q$.

Case 1: There is a good input. Suppose that there exists a good input $x_0^\star \in [k]$ such that $\{x_0^\star\} \times [a] \subseteq P_q$ for some q. Let $\mathcal{L}_{x_0^\star}$ be the set of leaves of π that contain an input where $x_0 = x_0^\star$, that is,

$$\mathcal{L}_{x_0^\star} = \{\ell \in \mathcal{L} : \exists (x_1, y_0, y_1, z, i) \text{ with } \pi(x_0^*, x_1; y_0, y_1, z, i) = \ell\}$$

and let $\overline{\mathcal{L}_{x_0^\star}}$ denote the complementary set of leaves, that is $\overline{\mathcal{L}_{x_0^\star}} = \mathcal{L} \setminus \mathcal{L}_{x_0^\star}$. We will show that $|\mathcal{L}_{x_0^\star}| \geq L_d^B(f)$ and that $|\overline{\mathcal{L}_{x_0^\star}}| \geq 2k - 2$. As a result, we get that

$$L(\pi) \geq |\mathcal{L}_{x_0^\star}| + \overline{\mathcal{L}_{x_0^\star}} \geq L_d^B(f) + 2k - 2,$$

as desired. First, we show that $|\mathcal{L}_{x_0^\star}| \geq L_d^B(f)$. Let π' be the d-round Bob-first subprotocol of π given when Alice says that her input is in P_q at the first round. Since $\{x_0^\star\} \times [a] \subseteq P_q$, it follows that π' computes $f(x_1, y_1)$ on input $(x_0^\star, x_1; y_0, y_1, 0, 1)$ for all $x_1 \in [a]$ and $y_1 \in [b]$. This yields a d-round Bob-first protocol for computing f, and therefore, the number of leaves in π' containing the input $x_0^\star$ must be at least $L_d^B(f)$. Hence, $|\mathcal{L}_{x_0^\star}| \geq L_d^B(f)$. Next, we argue that $|\overline{\mathcal{L}_{x_0^\star}}| \geq 2k - 2$. Consider the set of leaves given by

$$\{\pi(x_0, x_1; y_0, y_1, z, i) : i = 0, x_0 = y_0 \in [k] \setminus \{x_0^\star\}, y_1 = 1, z \in \{0, 1\}\}.$$

If we consider the restriction of π to the inputs $y_1 = 1$ and $i = 0$, we can apply Lemma 20 to conclude that all $2k - 2$ leaves in this set are in $\overline{\mathcal{L}_{x_0^\star}}$ and are pairwise distinct.

Case 2: No good input. Now we consider the case where there is no good input. For each $x_0 \in [k]$, we define a set $A_{x_0} \subseteq [a]$ of cardinality 2 as follows. Since $x_0 \in [k]$ is not good, there exists a pair $(x_1, x_1') \in [a]^2$ such that $\pi(x_0, x_1; y_0, y_1, z, i) \neq \pi(x_0, x_1'; y_0, y_1, z, i)$ for all y_0, y_1, z and i. Let $A_{x_0} = \{x_1, x_1'\}$. This completes our definition of A_{x_0}. We claim that the $4k$ inputs in the following set are all in pairwise distinct leaves:

$$W = \{(x_0, x_1, y_0, y_1, z, i) : x_0 = y_0 \in [k], i = 0, x_1 \in A_{x_0}, y_1 = 1, z \in \{0, 1\}\}.$$

To see this, suppose that $w \neq w'$ for some $w, w' \in W$. Let $w = (x_0, x_1; y_0, y_1, z, i)$ and $w' = (x_0', x_1'; y_0', y_1', z', i')$. We prove $\pi(w) \neq \pi(w')$ by considering the following three cases.

1. If $x_0 = x_0'$ and $x_1 \neq x_1'$, then we know that $\{x_1, x_1'\} = A_{x_0}$. By the construction of A_{x_0}, we can conclude that $\pi(w) \neq \pi(w')$.

2. If $x_0 = x_0'$ and $x_1 = x_1'$, then we must have $x_0 = y_0 = x_0'$ and $z \neq z'$ since $w \neq w'$; using Lemma 20, we conclude that $\pi(w) \neq \pi(w')$.

3. Lastly, suppose that $x_0 \neq x_0'$. If we consider the restriction of π to the inputs $y_1 = 1$ and $i = 0$, we can apply Lemma 20 to conclude that $\pi(w) \neq \pi(w')$.

□

Using this lower bound, we show one can approximately compute round-d complexity given an oracle that approximately computes round-$(d+1)$ complexity. We consider the following notion of approximation.

Definition 22. *For every constant $\epsilon > 0$, we say that an oracle $\mathcal{O}$ is a $(1+\epsilon)$-approximation of a function $\mathcal{L}(\cdot)$ if there exists a constant c such that, for all g in the domain of $\mathcal{L}(\cdot)$,*

$$\mathcal{L}(g) \leq \mathcal{O}(g) \leq (1 + \epsilon) \cdot \mathcal{L}(g) + c.$$

Corollary 23. *Let $0 < \epsilon < \frac{1}{8}$. Given an oracle computing a $(1+\epsilon)$-approximation of $L_{d+1}^A(\cdot)$ and a function $f : [a] \times [b] \to \{0, 1\}$, one can deterministically compute a $(1 + 4\epsilon)$-approximation of $L_d^B(f)$ in time $(ab)^{O(1)}$.*

Proof. First, we give the reduction algorithm and then we prove its correctness. Suppose $\mathcal{O}$ is an oracle that computes an approximation of $L^A_{d+1}(\cdot)$ satisifying, for all functions g,

$$L^A_{d+1}(g) \leq \mathcal{O}(g) \leq (1+\epsilon)L^A_{d+1}(g) + O(1).$$

For a positive integer k, let $F_k : ([k] \times [a]) \times ([k] \times [b] \times \{0,1\} \times \{0,1\}) \to \{0,1\}$ be given by

$$F_k(x_0, x_1; y_0, y_1, z, i) = \begin{cases} \mathrm{XorEq}_k(x_0; y_0, z) & \text{if } i = 0, \\ f(x_1; y_1) & \text{if } i = 1. \end{cases}$$

The reduction computes

$$v := \max\{\mathcal{O}(F_k) - 2(1+\epsilon)k : k \in [ab]\},$$

and outputs $v' := (v+2)/(1-2\epsilon)$. It is easy to see that this reduction runs in time $(ab)^{O(1)}$. To prove the correctness of the reduction, we claim that

$$L^B_d(f) \leq v' \leq (1+4\epsilon) \cdot L^B_d(f) + O(1)$$

for all functions f. From Lemma 21, we know that for all k

$$\begin{aligned} \mathcal{O}(F_k) - 2(1+\epsilon)k &\leq (1+\epsilon)L^A_{d+1}(F_k) - 2(1+\epsilon)k + O(1) \\ &\leq (1+\epsilon)L^B_d(f) + O(1) \end{aligned}$$

so we have that

$$v' = \frac{v+2}{1-2\epsilon} \leq \frac{1+\epsilon}{1-2\epsilon} \cdot L^B_d(f) + O(1) \leq (1+4\epsilon) \cdot L^B_d(f) + O(1),$$

where the last inequality holds because $\epsilon < 1/8$. On the other hand, if $k = L^B_d(f)$, we have from Lemma 21 that

$$\begin{aligned} \mathcal{O}(F_k) - 2(1+\epsilon)k &\geq L^A_{d+1}(F_k) - 2(1+\epsilon)k \\ &\geq \min\{4k, 2k - 2 + L^B_d(f)\} - 2(1+\epsilon)k \\ &= \min\{4L^B_d(f), 3L^B_d(f) - 2\} - 2(1+\epsilon)L^B_d(f) \\ &= 3L^B_d(f) - 2 - 2(1+\epsilon)L^B_d(f) \\ &\geq (1-2\epsilon)L^B_d(F_k) - 2. \end{aligned}$$

Since $k = L^B_d(f) \leq ab$, we conclude that $v' = (v+2)/(1-2\epsilon) \geq L^B_d(F_k)$. □

Combining Corollary 23 with the hardness result for the $d = 3$ case in Theorem 18, we get that computing L_d is NP-hard (under a polynomial-time truth-table reduction).

Corollary 24. *For any integer $d \geq 3$, there exists an $\epsilon > 0$ such that given access to an oracle that computes a $(1+\epsilon)$-approximation of L^A_d, one can compute any language in* NP *in polynomial time.*

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